

EVOLUTION

Like all living things, animals undergo changes as each new generation succeeds the one before. These changes are usually so slight that they are very difficult to see, but over thousands or millions of years, they can completely alter the way animals look and also the way they behave. This process of change is called evolution. It allows animals to exploit new opportunities and to adapt to changes that take place in the world around them. Evolution works by modifying existing characteristics, usually through a series of extremely small mutations. The result of this is that every animal is a living store of evolutionary history—one that helps to show how different species are related.

Animal adaptation

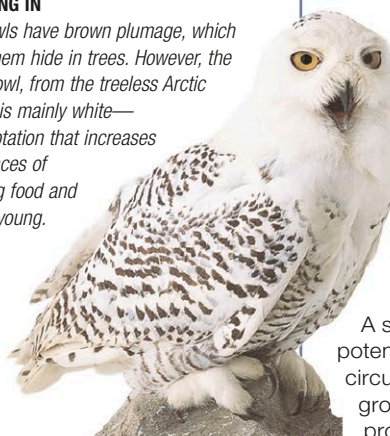
Evolution is made possible by the variations that exist within animals, and it occurs mainly because animals compete with each other for limited resources, such as space and food. In this competition, some characteristics prove to be more useful than others, which means that their owners are more likely to thrive, and to produce the most young. Animals with less useful features

Natural selection operates all the time, invisibly screening all the subtle variations that come about when animals reproduce. For example, for many animals, camouflage is a valuable aid to survival. Natural selection ensures that any improvements in an animal's camouflage—a slight change in color, pattern, or behavior—are passed on to the next generation, increasing its chances of survival and therefore its chances of producing young.

Adaptations such as camouflage last only as long as they are useful: if an animal's lifestyle changes, the path that evolution follows changes, too. This has happened many times with birds: some lineages have evolved the power of flight only to lose it when they take up life on land.



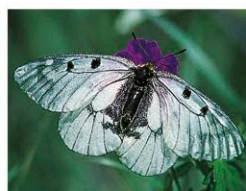
TAWNY OWL



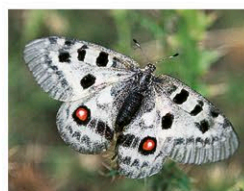
SNOWY OWL

Species and speciation

A species is any group of animals that has the potential to interbreed and that, under normal circumstances, does not breed with any other group. Speciation is one of the evolutionary processes that brings about new species. It usually occurs when an existing species



GERMAN FORM



.JAPANESE FORM

SPECIATION

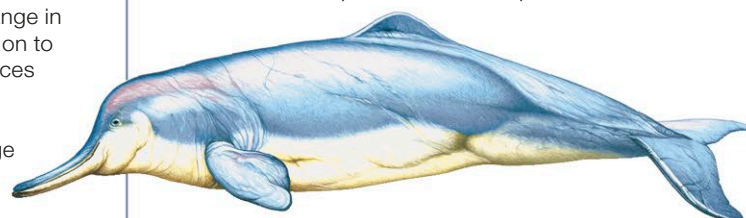
The Apollo butterfly is a highly variable species. Many forms are restricted to specific parts of the Northern Hemisphere. Since the Apollo often lives on mountains, its different forms tend to remain apart. In time, these forms could evolve into new species.

becomes split up into 2 or more isolated groups, which are kept apart either by physical barriers, such as seas and mountain ranges, or by changes in behavior. If these groups remain separate for long enough, they evolve their own characteristic adaptations and become so different that they can no longer interbreed.

Speciation is difficult to observe because it occurs so slowly, but evidence of it is not hard to find. Many animals—from butterflies to freshwater fish—show distinct regional differences. In time, these local forms, or subspecies, can become species in their own right.

Extinction

Extinction is a natural feature of evolution because for some species to succeed, others must fail. Since life began, about 99 percent of the Earth's species have disappeared and, on at least 5 occasions, huge numbers have died out in a relatively short time. The most recent of these mass extinctions, about 65 million years ago, swept away the dinosaurs and many other forms of life. However, despite such catastrophes, the



GOING, GOING, GONE

Declared extinct in August 2007, the Chinese river dolphin lived only in the Yangtze river basin. Increasing human activity caused the population to decline drastically from 400 in the 1980s to 100 in the mid-1990s. None has been found in recent surveys.

total number of living species has, until recently, followed a generally upward trend. Today, the extinction rate is increasing rapidly as a result of human interference in natural ecosystems. Primates, tropical birds, and many amphibians are particularly threatened. For the foreseeable future, this decline is set to continue because evolution generates new species far more slowly than the current rate of extinction.

Convergence

Unrelated animals often develop very noticeable similarities. For example, sharks and dolphins are fundamentally very different, but both have streamlined bodies with an upright dorsal fin—a shape that gives them speed and stability underwater—while moles and marsupial moles share a range of adaptations for life underground. Amphisbaenians and caecilians also look very similar, although the former are reptiles and the latter are highly unusual amphibians. Such similarities are the result of convergent evolution.

face more difficulties and find it harder to breed. The least successful animals are therefore gradually weeded out, while those with “winning” characteristics become widespread. This weeding-out process is called natural selection.

A HISTORY OF LIFE

Geologists divide earth's history into periods characterized by major physical changes, such as bursts of volcanic activity, collisions between continents, or alterations in climate. Many of these periods have ended in worldwide mass extinctions. For each period described below, the date shown indicates the period's end.

NUMBER OF FAMILIES	PRECAMBRIAN		CAMBRIAN		ORDOVICIAN		SILURIAN		DEVONIAN		CARBONIFEROUS	
	800	This vast expanse of geological time stretches from when the continents first formed to when animals with hard body parts first appeared in the fossil record. Life emerged near the beginning of Archean Eon, more than 4 billion years ago. The date for the first animals is less certain: being soft-bodied, they left few traces, although their existence is evident from fossils of burrows and tracks (about one billion years old).	The start of the Cambrian period was marked by an extraordinary explosion of animal life in the seas. Cambrian animals included mollusks, echinoderms, and arthropods. These were among the first creatures to have hard, easily fossilized body parts. By the end of this period, all the major divisions of animal life, or phyla, that exist today had been established.	During this period, life was still confined to the sea. Animals included the earliest crustaceans and some of the earliest jawless fish. Trilobites (arthropods with a 3-lobed body) were numerous, as were long-shelled nautiloids (predatory mollusks with sucker-bearing arms). Like the Cambrian period, the Ordovician ended in a mass extinction, probably caused by climatic changes.	Silurian times saw the evolution of the first fish with jaws and of giant sea scorpions—relatives of today's arachnids. The first land plants appeared.	In Devonian times, jawed and jawless fish diversified rapidly, which is why the period is known as the "Age of Fishes." Many fish lived in freshwater, where warm conditions and falling water levels encouraged the evolution of primitive lungs. As a result, amphibians evolved, becoming the first vertebrates to live on land. On land, insects became widespread, and the first true forests began to form. The Devonian ended with the third mass extinction, which killed up to 70 percent of animal species.	During this period, which is also called the "Coal Age," a warm global climate encouraged the growth of forests on swampy ground—home to amphibians and flying insects, including dragonflies with wingspans of up to 23½ in (60 cm). In the sea, ammonoids (mollusks related to today's nautiluses), were common.					
	700											
	600											
	500											
	400											
	300											
	200											
	100											
	0											
YEARS AGO (MILLIONS):		542	488	444	416	359	299					
PALAEOZOIC ERA												